Analytics and Metrics: quick notes

DEFINITIONS

- **Metric:** a quantifiable measure of success (or failure) against a goal.
- **KPI:** the metrics that give the most insight into the goal.
- **Anatomy:** definition, value, time grain, cohort. Always state all four.
- **North Star Metric (NSM):** the one metric that signifies value delivered (for example Swiggy: successful orders delivered; Airbnb: bookings and successful stays).
- **Input metric:** something the team controls. **Output metric:** the result the input causes.
- **Leading metric:** an early signal of future success. **Lagging metric:** a confirmed past result.
- **Guardrail (counter) metric:** protects against side effects of the primary metric.

THE MENTAL MODEL (USE IT FOR ANY "DEFINE SUCCESS" QUESTION)

CUSTOMER

1. Identify all users and stakeholders.
2. Identify each Job To Be Done (JTBD).
3. Define a metric that captures JTBD completion.

BUSINESS

4. Identify the business goal (shareholder value, vision, mission).
5. Define a metric that shows the product supports it.
6. Add guardrails for each primary metric.

METRIC SYSTEM: FIVE LAYERS

Layer	Question	Example (B2B collaboration)
North Star	What long-term value are we creating?	Weekly active teams
Input	What behaviours drive the result?	Invites sent, workflows created
Output	What moved after launch?	Activation rate, workflow completion
Guardrail	What should not get worse?	Error rate, latency
Diagnostic	Where exactly is the problem?	Drop-off after the invite step

THE 2×2

	Predictive (leading)	Retrospective (lagging)
High control	Leading inputs: Slack invites sent in onboarding; CRM integrations on Day 0	Lagging inputs: story points shipped; paid budget deployed (effort, not value)
Low control	Leading outputs: Slack's 2,000-message magic moment; Duolingo day-7 streak	Lagging outputs: ARR, net dollar retention, LTV, LTV:CAC, EBITDA

Frameworks and checks

AARRR (LIFECYCLE)

Stage	Question	Sample metrics
Acquisition	Are users showing interest?	Installs, site visits, CAC
Activation	Are users reaching the magic moment?	Active users, first-value event rate
Retention	Are users coming back?	D/W/M retention, churn
Referral	Are users bringing others?	NPS, viral coefficient, referral share
Revenue	Are users paying?	Sales, LTV (LTV must be much greater than CAC)

HEART (QUALITY OF EXPERIENCE)

Happiness, **E**ngagement, **A**doption, **R**etention, **T**ask success. Use with AARRR: AARRR tracks the funnel, HEART judges quality.

RETENTION AND PRODUCT-MARKET FIT

- A healthy curve **flattens** at a sustainable level (product A in the class).
- A curve that keeps falling in a straight line never flattens: no product-market fit yet (product B).
- Fix retention before spending on acquisition. Otherwise the bucket keeps leaking.

GROWTH LOOPS (COMPOUNDING, NOT LINEAR)

- **Viral / collaboration:** viral coefficient K above 1.0 (Figma, Slack, Miro, Notion).
- **Content / SEO (user-generated):** crawl-to-index ratio and UGC per active user (Stack Overflow, Quora, Pinterest).
- **Paid reinvestment:** LTV:CAC above 3.0 and CAC payback under 12 months (Duolingo, Uber).

GOODHART'S LAW AND GUARDRAILS

- "When a measure becomes a target, it ceases to be a good measure."
- Traps: notification trap (DAU up, uninstalls up), signup friction trap (low-intent churn), feature tour trap (clicks without learning).
- Defences: paired counter-metrics, permanent holdout (for example 5%), qualitative triangulation.
- **Senior PM rule:** never ship an experiment without its paired guardrail.

OKRS

- **Objective:** visionary, significant, inspirational direction.
- **Key result:** time-bound, measurable, verifiable (for example, cut disconnections by 30% in FY26).
- Cascade: company North Star, input drivers, team OKRs, initiatives and sprints.